

Request processing lifecycle (one postback)

1 Restore View

Postback: restore the component tree from the view state. Initial GET: create an empty UIViewRoot and jump to 6.

2 Apply Request Values

decode() stores each submitted value. immediate inputs convert + validate here; immediate commands run their action here.

3 Process Validations

Convert, then validate (required, f:validate*, Bean Validation). Any failure: message + valid=false → skip to 6.

4 Update Model Values

Call the setters of the value expressions, e.g. # {review.text}. Model untouched if 3 failed.

5 Invoke Application

actionListener then action; the outcome goes to the NavigationHandler. f:viewAction runs here.

6 Render Response

Build the view (initial), encode the tree to HTML or partial-response XML, save the state.

renderResponse() jumps to 6; responseComplete() ends the lifecycle (redirect, download). PhaseListener: getPhaseId(), beforePhase/afterPhase.

Ajax (partial lifecycle): phases 2–4 visit only the execute components, phase 6 encodes only render. **immediate command**: non-immediate inputs are not converted, validated or applied.

Pitfalls: logic in getters (called many times), immediate to "skip validation" (scope Ajax execute instead), binding to session beans, state-changing GET.

Setup, context parameters, project stages

Server: jakarta.platform:jakarta.jakartaee-web-api:11.0.0 provided (pulls jakarta.faces-api:4.1.2); no Faces JARs in the WAR. **Tomcat:** org.glassfish:jakarta.faces:4.1.17 or myfaces-apt/impl:4.1.4 + weld-servlet-shaded:6.0.4.Final.

WEB-INF/ web.xml (6.1), faces-config.xml (version="4.1"), beans.xml (bean-discovery-mode="annotated"; required on Tomcat + Weld). @FacesConfig on a CDI bean activates Faces when there is no descriptor.

FacesServlet mappings: *.xhtml (recommended), /faces/*, extensionless with AUTOMATIC_EXTENSIONLESS_MAPPING (4.0).

jakarta.faces...	default
PROJECT_STAGE Development	Production
UnitTest SystemTest Production	
FACELETS_REFRESH_PERIOD (s; -1 = never)	0; -1 in Prod.
FACELETS_SKIP_COMMENTS	false
STATE_SAVING_METHOD server client	server
INTERPRET_EMPTY_STRING_SUBMITTED_VALUES_AS_NULL	false
VALIDATE_EMPTY_FIELDS	auto
DATETIMECONVERTER_DEFAULT_TIMEZONE_IS_SYSTEM_TIMEZONE	false (GMT)
CLIENT_WINDOW_MODE none url	none
ENABLE_WEBSOCKET_ENDPOINT	false
validator.ENABLE_VALIDATE_WHOLE_BEAN	false

Misspelt or old javax.faces.* names are **ignored silently**. Vendor extras: com.sun.faces.*, org.apache.myfaces.*.

Facelets, templating, EL

prefix	namespace	URN
h	jakarta.faces.html	
f	jakarta.faces.core	
ui	jakarta.faces.facelets	
cc	jakarta.faces.composite	
p/pt attrs	jakarta.faces.passthrough	
faces elements	jakarta.faces (faces:id)	
c / fn	jakarta.tags.core / jakarta.tags.functions	

Pages: <!DOCTYPE html>, well-formed XHTML, h:head + h:body (needed for resources), f:metadata first.

Templates

Template /WEB-INF/templates/layout.xhtml: <ui:insert name="content">default</ui:insert> Client: <ui:composition template="/WEB-INF/templates/layout.xhtml"> + <ui:define name="content">. Also ui:include src + ui:param, ui:decorate, ui:fragment rendered, ui:repeat value var varStatus, ui:remove, ui:debug. A misspelt ui:define name is ignored silently.

JSTL vs components

c:if, c:forEach, c:set run at **view build time**; rendered, ui:repeat at **render time**. Never use c:if on a row variable of ui:repeat/h:dataTable.

Expression Language 6.0

Value #{catalogue.books} (get/set) · method # {review.save}, #{catalogue.remove(book)} · \${...} acts like #{...} in Facelets. Operators empty eq ne lt gt le ge and or not ? : += (concat), lambdas and streams # {books.stream().filter(b -> b.year gt 2000).toList()}. Implicit objects: facesContext view component cc request session application param paramValues header cookie initParam flash resource requestScope viewScope sessionScope applicationScope flowScope.

h: and f: tags, conversion, validation, i18n

Inputs h:inputText inputSecret inputTextarea inputHidden, inputFile (Part; enctype="multipart/form-data") · **select** selectOneMenu selectOneRadio selectManyCheckbox selectBooleanCheckbox + f:selectItems value var itemLabel itemValue.

Output h:outputText (escapes), outputFormat, outputLabel for, graphicImage name library, outputStyleSheet, outputScript · **POST** commandButton/commandLink in h:form · **GET** h:button/h:Link outcome · h:dataTable value var + h:column + f:facet name="header".

Converters

Implicit by type (numbers, enums, UUID 4.1) · f:convertNumber type="currency" · f:convertDateTime type="localDate" pattern="yyyy-MM-dd" · @FacesConverter(value="isbn", managed=true) + Converter<Isbn>: getAsObject/getAsString, throw ConverterException.

Validators

required="true" requiredMessage · f:validateLength, validateLongRange, validateDoubleRange, validateRegex, validateRequired · @FacesValidator(managed=true) → ValidatorException.

Bean Validation is automatic when on the classpath: @NotBlank @Size @Min on the model; f:validateBean validationGroups disabled; class-level f:validateWholeBean (needs the context parameter).

Messages and i18n

h:message for, h:messages globalOnly; addMessage(clientIdOrNull, new FacesMessage(SEVERITY_ERROR, summary, detail)); label names the field. faces-config.xml: resource-bundle (base-name + var="msg"), message-bundle overrides standard texts, locale-config en + es; f:view locale.

Events, navigation, view params, PRG

Events: ActionEvent (actionListener, f:setPropertyActionListener), ValueChangeEvent, PhaseEvent, system events via f:event type="postAddToView | preValidate | postValidate | preRenderView".

Outcomes: action="details" → details.xhtml (implicit); null/void = stay on the view; "details?faces-redirect=true&includeViewParams=true" = redirect. Rules: navigation-rule / navigation-case with from-outcome, if, to-view-id, redirect.

Bookmarkable:

```
<f:metadata>
<f:viewParam name="id" value="#{bookPage.id}" required="true"/>
<f:viewAction action="#{bookPage.load}"/>
</f:metadata>
```

viewParam converts/validates the query parameter in phases 2–4; viewAction runs in phase 5 on GET (onPostback="false", phase, if) and may return an outcome.

Post/Redirect/Get: save, then return "confirmation?faces-redirect=true"; with facesContext.getExternalContext().getFlash().setKeepMessages(true); read #{flash.title} once on the next view.

CDI beans, scopes, Faces Flows

Backing bean = CDI bean: @Named @RequestScoped public class CatalogueBean (EL name catalogueBean); @Inject services, FacesContext, ExternalContext, Flash, @ManagedProperty("#{param.q}") (jakarta.faces.annotation), @PostConstruct for set-up (not getters).

scope	lives for
@RequestScoped	one HTTP request
@ViewScoped (jakarta.faces.view; Serializable)	postbacks to the same view
@FlowScoped("borrow")	while inside the flow
@ClientWindowScoped (4.0, jakarta.faces.lifecycle)	one browser tab; needs CLIENT_WINDOW_MODE=url
@ConversationScoped	begin() ... end()
@SessionScoped / @ApplicationScoped	user session / application
@Dependent	its injection target

Keep page state out of session scope and @ViewScoped beans small.

Faces Flows

Directory /borrow/ with borrow-flow.xml, or @Produces @FlowDefinition Flow define(@FlowBuilderParameter FlowBuilder fb). Nodes: view, switch, method call, return, flow call; inbound/outbound parameters; # {flowScope}; enter with action="borrow".

Ajax, search expressions, WebSocket push

<h:inputText value="#{search.q}"> <f:ajax event="keyup" delay="300" execute="@this" render="results msgs" listener="@{search.run}"/> </h:inputText>

Defaults: execute="@this", render="@none"; default event = action (commands) or valueChange (inputs). Also onerror, onerror, resetValues, disabled, immediate. Script faces.js: faces.ajax.request(source, event, {execute, render}); faces.ajax.addOnError.

Search keywords: @this @form @all @none @parent @namingcontainer @next @previous @child(n) @id(id) @root @composite; ids are relative to the nearest naming container (h:form, tables, composites); a leading : starts from the view root; client id separator :.

Push: ENABLE_WEBSOCKET_ENDPOINT=true; view <f:websocket channel="reviews" onmessage="showReview" scope="application"/> (also session, view, user, onopen onerror onClose, nested f:ajax); bean @Inject @Push(channel="reviews") PushContext reviews; reviews.send(json);. Mojarra on Tomcat also needs a JSON-P implementation (Parsson).

Composite and custom components

Composite: resources/bookshelf/rating.xhtml → xmlns:bs="jakarta.faces.composite/bookshelf" → <bs:rating value="#{review.stars}"/>. cc:interface: cc:attribute (type default required method-signature targets), cc:editableValueHolder, cc:clientBehavior; cc:implementation uses #{cc.attrs.value}; Java backing via componentType.

Tag files: *.taglib.xml tag + source, registered by FACELETS_LIBRARIES or META-INF.

Custom component: extend UIComponentBase/UIInput, @FacesComponent(value, createTag=true, tagName, namespace), keep attributes in getStateHelper(). **Renderer:** @FacesRenderer, decode, encodeBegin/encodeEnd with ResponseWriter. **Behaviors:** @FacesBehavior + ClientBehaviorBase, getScript().

State saving, exceptions, security

State: hidden jakarta.faces.ViewState; server (session, limited number of views) or client (set an encryption key in a cluster); stateless view f:view transient="true"; serialisable beans for clustering.

Exceptions: web.xml <error-page><exception-type>jakarta.faces.application.ViewExpiredException</exception-type> custom ExceptionHandlerWrapper + ExceptionHandlerFactory; Ajax errors arrive as <error> in the partial response.

Security: Jakarta Security 4.0 @CustomFormAuthenticationMechanismDefinition n + SecurityContext.authenticate(), identity store; <security-constraint> for URLs; rendered="# {request.isUserInRole('LIBRARIAN')}" hides UI only, re-check in services. CSRF: postbacks carry the view state; protect GET entry views with <protected-views>. XSS: never escape="false" on user input. CSP: nonce support in Mojarra 4.0.16+ (com.sun.faces.enableCspNonce), standard support planned for 5.0.

PrimeFaces, OmniFaces, runtimes, testing

PrimeFaces 16 xmlns:p="primefaces": p:ajax, p:commandButton process update (search expressions plus @widgetVar(x), @(.css)); p:dataTable lazy="true" + LazyDataModel/JPAJazyDataModel, paginator, sort, filter, selection, editing, export; p:dialog, dialog framework; PrimeFaces.current().ajax().update(...); themes primefaces.THEME (saga, vela, aria). **Extensions** xmlns:pe="primefaces.extensions".

OmniFaces 5 xmlns:of="omnifaces": Faces/Messages, @Param, @Eager, o:validateEqual, SelectItemsConverter, FullAjaxExceptionHandler, CombinedResourceHandler.

Runtimes: GlassFish 8, Payara 7 and WildFly 41 bundle Mojarra; Open Liberty (faces-4.1) and TomEE 10 (Faces 4.0) bundle MyFaces; Tomcat 11 or Jetty 12 + Weld 6: implementation in WEB-INF/lib; Spring Boot 4.1 + JoinFaces 6.1 (joinfaces.faces.*); Quarkus + myfaces-quarkus (views in META-INF/resources).

Testing: JUnit 5 + Mockito for beans, converters, validators; MyFaces Test mocks; Arquillian (in-container, client mode); Selenium, Playwright for Java, HtmlUnit; PrimeFaces Selenium page objects.

Changed since older tutorials and books (JSF 1.x–2.2, PrimeFaces 4/5)

JSP views deprecated 2.0, removed 4.0 → **Facelets XHTML**

jsf.js, jsf.ajax → faces.js, faces.ajax (4.0)

@ManagedBean, @ManagedProperty (jakarta.faces.bean) deprecated 2.3, removed 4.0 → **CDI @Named + scopes**

RequestContext (PrimeFaces ≤ 6.x) → PrimeFaces.current() (6.2/7.0)

javax.faces.* → jakarta.faces.* in Faces 3.0 (Jakarta EE 9)

MethodBinding, ValueBinding, ResourceResolver removed 4.0 → **EL MethodExpression / ValueExpression, ResourceHandler**

xmlns:jcp.org/jsf/*, java.sun.com/jsf/* → jakarta.faces.* URNs (4.0; old URIs removal planned for 5.0)

Full state saving deprecated 4.1; **ActionSource2** deprecated 4.1; Faces 5.0 (EE 12, Java 21) is upcoming